

Def. Sia E misurabile $\subseteq \mathbb{R}^n$

$$L^\infty(E) := \left\{ f: E \rightarrow \mathbb{R} \mid \text{ess-sup}_{x \in E} |f(x)| < +\infty \right\}$$

$\uparrow$
essenzialmente limitate

$$\text{ess-sup}_{x \in E} |f(x)| := \min \{ M: |f(x)| \leq M \text{ q.o. } x \in E \}$$

$$\|f\|_\infty := \text{ess-sup}_{x \in E} |f(x)|.$$

Esempio $E = \mathbb{R}$

$$f(x) = \begin{cases} 0 & x \notin \mathbb{N} \\ x & x \in \mathbb{N} \end{cases}$$

f non limitata su $\mathbb{R}$: $f\left(\underbrace{n}_{\in \mathbb{N}}\right) = n \rightarrow +\infty$

f essenzialmente limitata: $|\mathbb{N}| = 0. \quad \|f\|_\infty = 0.$

$(L^\infty(E), \|\cdot\|_\infty)$ sp. lett. normato

Teorema $(L^\infty(E), \|\cdot\|_\infty)$ è un Banach.

$$\left[f \in L^p(E) \quad \forall p \geq \bar{p}, \quad \lim_{p \rightarrow +\infty} \|f\|_p = \|f\|_\infty \right]$$

Appartenenza a $L^\infty(E)$

$$f \in L^\infty(E) \iff \text{ess-sup}_{x \in E} |f(x)| < +\infty.$$

Convergenza in $L^\infty(E)$

$$f_n \rightarrow f \text{ in } L^\infty(E) \iff \text{ess-sup}_{x \in E} |f_n - f| \xrightarrow{n \rightarrow +\infty} 0$$

Confronto tra spazi $L^p(E)$

$$p < q \quad (p, q \in [1, +\infty]) \quad \begin{matrix} ? \\ \Rightarrow \\ ? \end{matrix} \quad \begin{matrix} L^p(E) \subseteq L^q(E) \\ L^q(E) \subseteq L^p(E) \end{matrix}$$

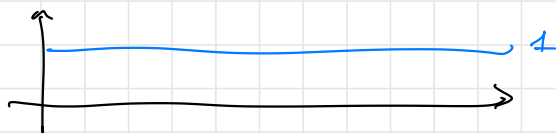
In generale NO

$$L^1(0, +\infty)$$

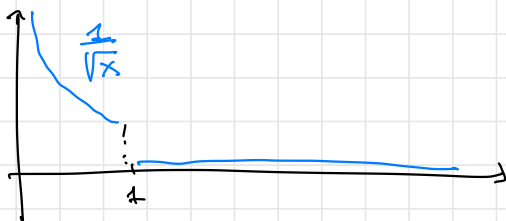
$$L^2(0, +\infty)$$

$$L^\infty(0, +\infty) \subset \mathbb{R}_+$$

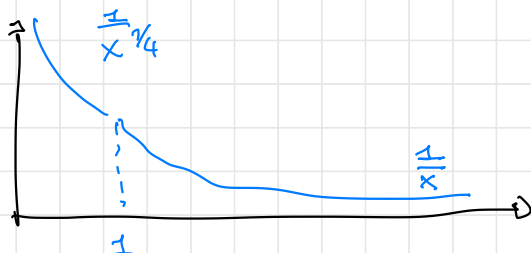
- $f \in L^\infty(\mathbb{R}_+)$ ma $f \notin L^1(\mathbb{R}_+)$, $f \notin L^2(\mathbb{R}_+)$



- $f \in L^1(\mathbb{R}_+)$ ma $f \notin L^2(\mathbb{R}_+)$, $f \notin L^\infty(\mathbb{R}_+)$



- $f \in L^2(\mathbb{R}_+)$ ma $f \notin L^1(\mathbb{R}_+)$, $f \notin L^\infty(\mathbb{R}_+)$



Disuguaglianza di Hölder

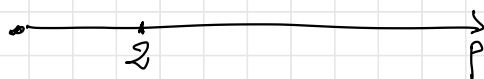
Sia $E \text{ mis.} \subseteq \mathbb{R}^1$, $p \in [1, +\infty]$.

Sia $\boxed{p' := \frac{p}{p-1}}$ con le convenzioni $\frac{1}{\infty} = 0$
 $\frac{\infty}{\infty} = 1$
 $\uparrow$
esponente coniugato

(equivalentemente $\frac{1}{p'} + \frac{1}{p} = \frac{p-1}{p} + \frac{1}{p} = \frac{p}{p} = 1$)

$$\begin{aligned} 1' &= \infty \\ \infty' &= 1 \end{aligned}$$

$$2' = 2$$



Sia $f \in L^p(E)$ e sia $g \in L^{p'}(E)$

Allora $f \cdot g \in L^1(E)$ e vale:

$$\boxed{\|f \cdot g\|_1 \leq \|f\|_p \cdot \|g\|_{p'}}$$

Esempi

- $f, g \in L^2(E) \Rightarrow \|f \cdot g\|_1 \leq \|f\|_2 \|g\|_2$
- $f \in L^1(E), g \in L^\infty(E) \Rightarrow \|f \cdot g\|_1 \leq \|f\|_1 \|g\|_\infty$

$$\int_E |f(x) g(x)| \leq \left(\sup_{x \in E} |g(x)| \right) \cdot \int_E |f(x)|$$

$\uparrow$
 $|f(x) g(x)| \leq |f(x)| \cdot \sup_{x \in E} |g(x)|$
q.o. su E .

Conseguenze d. Hölder

1) PROPRIETÀ DI INNESTO $L^q(E) \subseteq L^p(E)$

Sia $E \subseteq \mathbb{R}^n$ con $m(E) = |E| < +\infty$

Siano $p, q \in [1, +\infty]$, con $\boxed{p \leq q}$

Allora $f \in L^p(E)$, e vale

$$(*) \rightarrow \|f\|_p \leq m(E)^{\frac{q-p}{pq}} \|f\|_q \quad \forall f \in L^q(E).$$

(In particolare: $f \in L^\infty(E)$, $m(E) < +\infty \Rightarrow f \in L^p(E) \quad \forall p \in [1, +\infty]$)

Dim. Sia $f \in L^q(E)$ con $|E| < +\infty$, $q \geq p$.

$$\int_E (|f|^p \cdot 1) = \int_E \underbrace{(|f|^p)}_{\in L^{q/p}} \cdot \underbrace{\chi_E}_{\in L^{(q/p)'}}$$

$$\cdot |f|^p \in L^{\frac{q}{p}}(E) \text{ perché } \int (|f|^p)^{\frac{q}{p}} = \int |f|^q < +\infty$$

$$\cdot \chi_E \in L^{\left(\frac{q}{p}\right)'}(E) \text{ perché } \int_E |\chi_E|^{\left(\frac{q}{p}\right)'} = |E| < +\infty$$

Hölder:

$$\int_E |f|^p \leq \underbrace{\| |f|^p \|_{q/p}}_{(1)} \cdot \underbrace{\| \chi_E \|_{(q/p)'}}_{(2)}$$

$$(1) \quad \| |f|^p \|_{q/p} = \left[\int_E |f|^p |^q \right]^{p/q} = \left[\int_E |f|^q \right]^{p/q}$$

$$(2) \quad \| \chi_E \|_{(q/p)'} = \left[\int_E \underbrace{|\chi_E|^{(q/p)'}}_1 \right]^{\frac{1}{(q/p)'}} =$$

$$= |E|^{\frac{1}{(q/p)'}} = |E|^{\frac{q-p}{q}}$$

$$\frac{1}{(q/p)'} = 1 - \frac{1}{(q/p)} = 1 - \frac{p}{q} = \frac{q-p}{q}$$

$$\int_E |f|^p \leq \| |f|^p \|_{q/p} \cdot \| \chi_E \|_{(q/p)'}$$

$$= \left(\int_E |f|^q \right)^{p/q} \cdot \left(|E|^{\frac{q-p}{q}} \right)$$

Elevo tutto alla $\frac{1}{p}$:

$$\left(\int_E |f|^p \right)^{\frac{1}{p}} \leq |E|^{\frac{q-p}{pq}} \left(\int_E |f|^q \right)^{\frac{1}{q}}$$

$$\|f\|_p \leq \|f\|_q$$

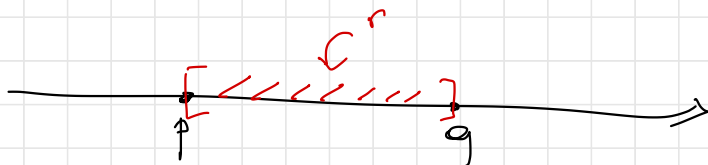
□

2) PROPRIETÀ DI INTERPOLAZIONE

Sia $E \text{ mis} \subseteq \mathbb{R}^n$, e siano $p < q$. ($p, q \in [1, +\infty]$)

Sia $f \in L^p(E) \cap L^q(E) \Rightarrow$

$f \in L^r(E) \quad \forall r \in [p, q]$



e vale:

$$\|f\|_r \leq \|f\|_p^\alpha \cdot \|f\|_q^{1-\alpha}$$

dove $\alpha \in (0, 1)$ e tale che

$$\frac{1}{r} = \alpha \cdot \frac{1}{p} + (1-\alpha) \cdot \frac{1}{q}$$

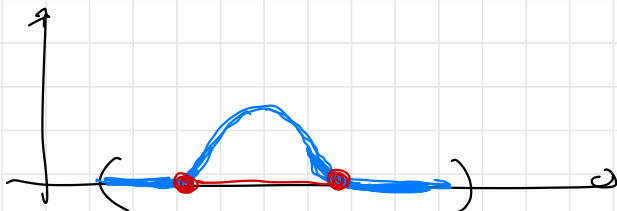
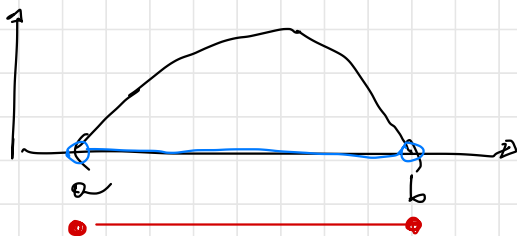
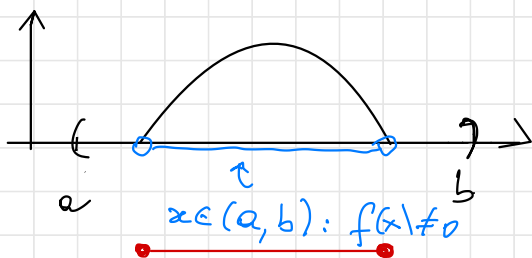
Teorema di approssimazione in L^p con funzioni "regolari"

Sia Ω aperto $\downarrow \mathbb{R}^n$, e sia $p \in [1, +\infty)$

Allora: $C_0^\infty(\Omega)$ è DENSE in $L^p(\Omega)$.

$:= \{f: \Omega \rightarrow \mathbb{R} \text{ di classe } C^\infty \text{ su } \Omega \text{ e aventi il supporto compatto in } \Omega\}$.

$$\text{supp } f := \overline{\{x \in \Omega : f(x) \neq 0\}} \quad (\text{chiuso})$$



DENSO significa che:

• $\forall f \in L^p(E) \quad \exists \{\varphi_n\} \subseteq C_0^\infty(\Omega)$ tale che

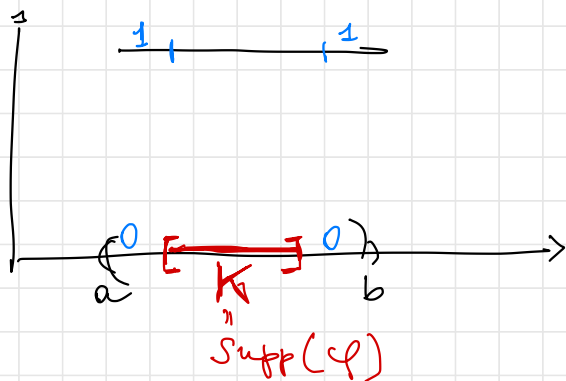
$$\varphi_n \rightarrow f \quad \text{in } L^p(E)$$

$$(\text{ovvero } \lim_n \|\varphi_n - f\|_p = 0)$$

• $\forall f \in L^p(E), \forall \varepsilon > 0, \exists \varphi \in C_0^\infty(E)$ tale che

$$\|f - \varphi\|_p < \varepsilon$$

FALSO per $p = +\infty$



$$\|f - \varphi\|_\infty \geq 1$$

$\uparrow \quad \quad \uparrow$
1 0

$$1 \equiv f \in L^\infty(a, b)$$

$\exists? \varphi \in C_0^\infty(a, b)$ tale che

$$\|f - \varphi\|_\infty < \varepsilon$$

NO

per $\varepsilon = \frac{1}{2}$

Prodotto di convoluzione

Dom. $\rightarrow$ Hölder : $f \in L^p(E), g \in L^p(E) \Rightarrow f \cdot g \in L^1(E)$

$\rightarrow f, g \in L^1(E) \not\Rightarrow f \cdot g \in L^1(E)$

esempio $E = (0, 1)$ $f(x) = g(x) = \frac{1}{\sqrt{x}}$
 $f, g \in L^1(0, 1)$ MA $f \cdot g = \frac{1}{x} \notin L^1(0, 1)$

Proposizione 1 Siano $f, g \in L^1(\mathbb{R})$. Posto

$$\underbrace{f * g}(x) \stackrel{\text{def}}{=} \int_{\mathbb{R}_y} f(x-y) \cdot g(y) dy,$$

$\uparrow$
PRODOTTO DI CONVOLOZIONE
TRA f e g

(i) $f * g(x)$ esiste (finito) per q.o. $x \in \mathbb{R}$

(ii) $f * g \in L^1(\mathbb{R}_x)$

(iii) $\|f * g\|_1 \leq \|f\|_1 \cdot \|g\|_1$

Osservazione:

- Vale la Proposizione 1 se $\mathbb{R} \mapsto \mathbb{R}^n$
- $f * g = g * f$
- $f \in L^1(\mathbb{R}), g \in L^p(\mathbb{R}) \Rightarrow \|f * g\|_p \leq \|f\|_1 \|g\|_p$

Dim.

$$(i) \quad H(x, y) := f(x-y)g(y) \quad (x, y) \in \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$$

Se sapessi $H \in L^1(\mathbb{R}^2)$, Fubini $\Rightarrow \exists f * g(x)$ finito per q.o. x

Considero $|H(x, y)| \geq 0$. Tonelli?

Verifico ip. Tonelli

$$\begin{aligned} \bullet \int_{\mathbb{R}_x} |H(x, y)| dx &= \int_{\mathbb{R}_x} |f(x-y)| |g(y)| dx = \\ &= |g(y)| \cdot \int_{\mathbb{R}_x} |f(x-y)| dx < +\infty \end{aligned}$$

$$\begin{aligned} \bullet \int_{\mathbb{R}_y} \left[\int_{\mathbb{R}_x} |H(x, y)| dx \right] dy &= \int_{\mathbb{R}_y} |g(y)| \cdot \|f\|_1 dy = \\ &= \|f\|_1 \cdot \underbrace{\int_{\mathbb{R}_y} |g(y)| dy}_{\|g\|_1} < +\infty \end{aligned}$$

$\Rightarrow$ Applico Tonelli: $|H| \in L^1(\mathbb{R}^2) \Rightarrow \underline{H \in L^1(\mathbb{R}^2)}$

Quindi posso applicare Fubini:

per q.o. $x \in \mathbb{R}$, $y \mapsto H(x, y) = f(x-y)g(y)$ appartiene a $L^1(\mathbb{R}_y)$

Adesso dim che vale (i).

(iii) Tesi: $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$

$$\boxed{\|f * g\|_1} = \int_{\mathbb{R}^n} |f * g|(x) dx = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y) g(y) dy \right| dx$$

$$\leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y) g(y)| dy dx$$

$$\stackrel{\substack{= \\ \uparrow \\ \text{Fubini}}}{=} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)| |g(y)| dx dy$$

$$= \int_{\mathbb{R}^n} |g(y)| \underbrace{\left(\int_{\mathbb{R}^n} |f(x-y)| dx \right)}_{\|f\|_1} dy = \int_{\mathbb{R}^n} |g(y)| \cdot \underbrace{\|f\|_1}_{\|f\|_1} dy$$

$$= \|f\|_1 \cdot \underbrace{\int_{\mathbb{R}^n} |g(y)| dy}_{\|g\|_1} = \boxed{\|f\|_1 \|g\|_1}$$

□

Proposizione 2. $f \in L^1(\mathbb{R}) \cap C^k(\mathbb{R})$, $g \in L^1(\mathbb{R})$
 $\text{Supp } f \text{ è compatto}$

Allora:

(i) $f * g \in C^k(\mathbb{R})$

(ii) $\forall \alpha \leq K \quad (f * g)^{(\alpha)} = f^{(\alpha)} * g$
 [Vale anche con $K = \infty$]

Idea dim.

$$f * g(x) = \int_{\mathbb{R}} f(x-y) g(y) dy$$

f continua $\Rightarrow f * g$ continua

$x_n \rightarrow x \Rightarrow f * g(x_n) \rightarrow f * g(x)$

$$f * g(x_n) = \int_{\mathbb{R}} \underbrace{f(x_n - y)}_{h_n(y)} g(y) dy$$

$h_n(y) := f(x_n - y) g(y)$

f continua $\Rightarrow \lim_{n \rightarrow \infty} h_n(y) = f(x - y) g(y)$

Cr. Dominata $\Rightarrow \int_{\mathbb{R}} h_n(y) dy \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} f(x-y) g(y) dy$ $\square$

[Om. Per avere $f * g$ ha supporto compatto,
 serve entrambe f e g abbiano supporto compatto.]

Idea della dm. di Fejér di densità ($\Omega = \mathbb{R}^{(n)}$)

Sia $n=1$, $p=1$.

Data $f \in L^1(\mathbb{R})$ voglio trovare $\{\varphi_n\} \subseteq C_0^\infty(\mathbb{R})$

tale che $\varphi_n \rightarrow f$ in $L^1(\mathbb{R})$.

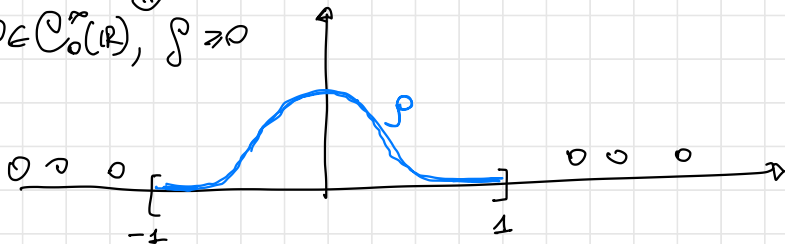
$$\varphi_n(x) \stackrel{\text{def}}{=} f * \underset{\substack{\uparrow \\ C_0^\infty(\mathbb{R})}}{\rho_n}(x) = \int_{\mathbb{R}} f(x-y) \rho_n(y) dy$$

successione di mollificatori

$\rho_n(x) \stackrel{\text{def}}{=} n \rho(nx)$, dove ρ è un nucleo di convoluzione.

• $\rho \in C_0^\infty(\mathbb{R}; \mathbb{R}^+)$ $\text{supp } \rho \subseteq [-1, 1]$, $\int_{\mathbb{R}} \rho(x) dx = 1$

$\rho \in C_0^\infty(\mathbb{R})$, $\rho \geq 0$



• $\rho_n \in C_0^\infty(\mathbb{R}; \mathbb{R}^+)$, $\text{supp } \rho_n \subseteq \left[-\frac{1}{n}, \frac{1}{n}\right]$, $\int_{\mathbb{R}} \rho_n(x) dx = 1$

[Ques. $\varphi_n = f * \rho_n$ sono C^∞ e convergono a f in L^1]
[T.A. non hanno il supporto compatto.]

$$f \cdot \chi_{[-1/k, 1/k]} = \underset{\substack{\uparrow \\ \text{Supp compatto}}}{f_k} \xrightarrow{K \rightarrow +\infty} f$$